Prism Bastion — Technical Sheet

Tech used

Single self-contained HTML file (about 24 KB), no libraries or external assets. HTML5 Canvas 2D with devicePixelRatio scaling (capped at 2x), neon glow via shadowBlur, an animated dashed light-road, rotating tower turrets, beams, splash rings, frost pulses, particle bursts, floating crystal numbers, screenshake and a twinkling starfield.

Web Audio API: all sound effects are synthesized (shots per tower type, explosions, leaks, upgrades, boss alarm), plus a scheduled procedural synthwave track that speeds up each wave and drops key during boss fights. Audio starts on first tap (iOS-safe) and suspends when the tab is hidden.

Pointer Events, 64px bottom-bar buttons and safe-area insets for notched phones. The board scales to fit any portrait screen. Auto-save and best score use localStorage.

Game systems

9x14 grid with a fixed winding road (8 turns). Enemies move by distance along the path, so targeting picks the raider furthest along.

4 tower types with 3 levels each (damage x1.75, range +0.35, fire rate x0.88 per level); single-target, armor-piercing, area slow and splash damage.

5 raider types with flat armor, slow effects and boss waves. 30 generated waves: health grows 20% per wave and the mix adds runners, swarms and tanks over time.

Wave-clear bonus, sell refunds at 70%, win bonus per life, save and continue.

Audit 1 — Balance (simulated play)

A headless simulator runs the same game code as the HTML file. Bots at 3 skill levels played 8 full games each with different random wave timing.

First pass: health grew 14.5% per wave and even a random Pulse-spamming bot nearly won, so the game had no challenge. At 21.5% no bot could win. Final tuning: 20% per wave, starting crystals cut from 160 to 140 and the wave bonus reduced to 15 + 2 per wave.

Final results: Novice (random Pulse spam): 0/8 won, median wave reached 21, range 19 to 21; Steady (planned mix, upgrades): 0/8 won, median wave reached 30, range 30 to 30; Expert (best spots, full mix): 8/8 won, median wave reached 30 (win), range 30 (win) to 30 (win). Weak play falls around wave 20, solid play reaches the final boss, and only well-placed, upgraded mixed defenses win.

Audit 2 — Bugs

Ran a headless smoke test of the real file with a stubbed browser: building, invalid taps on the road, not-enough-crystals, upgrading, selling, speed buttons, a full 30-wave win, the end screen, best score, saving, continuing and a forced loss. No errors.

Fixed: Frost slow strength was calculated after the slow timer was reset, so an old slow value could stick to an enemy. The new slow is now worked out first.

Fixed: the top bar and Wave button only refreshed while a wave was running, so they could show stale crystals after a wave ended. They now refresh continuously. Fixed: the low-lives warning did not reset between games.

Audit 3 — Player experience

First-time hints tell you to pick a tower, tap a tile and then press Start. Tapping the road or lacking crystals explains why nothing happened.

After each wave a toast shows the bonus and which raider types come next; boss waves get a warning and an alarm. A warning appears when 5 or fewer lives remain.

The tower panel shows exact damage, range, fire rate, kills, upgrade cost and refund, and the range circle is drawn on the board.